

WASP-36b

R-1985

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_150218 · created 2026-08-07T01:16:21+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457071.8422 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.065 +/- 0.036
Transit depth (Rp/Rs)^2	0.43 +/- 0.47 [%]
Orbital Inclination (inc)	83.29 +/- 2.82
Airmass coefficient 1 (a1)	1.201 +/- 0.012
Airmass coefficient 2 (a2)	-0.1107 +/- 0.0034
Scatter in the residuals of the lightcurve fit is	1.02 %
Best Comparison Star	None
Optimal Aperture	11.54
Optimal Annulus	30.64
Transit Duration (day)	0.062 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.065 ± 0.036 vs 0.1368 (deviation 1.99σ)
Duration fit vs published	0.062 d vs 0.0758 d (deviation 0.55σ)
Mid-time O–C	-2.7 min
Depth SNR	1.8
Residual scatter	1.02 %

Light curve

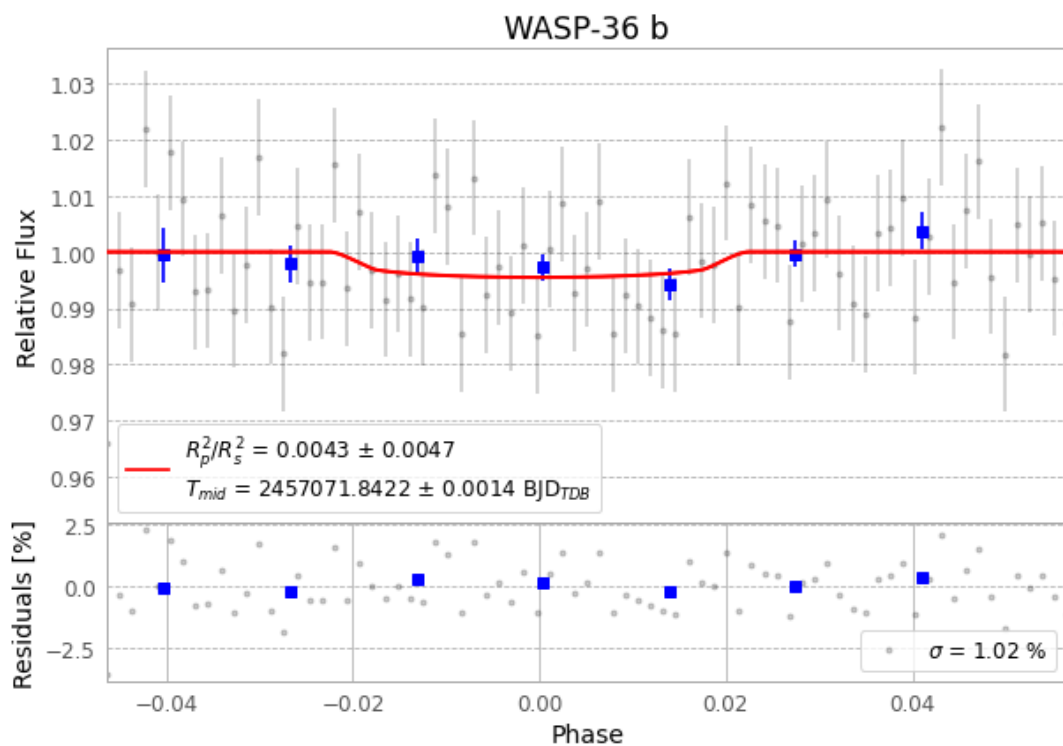


Figure 1 — FinalLightCurve_WASP-36 b_2015-02-17.png (SHA-256 67fdf4da875a3cdb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [355, 395], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 597306ca36f60cb3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

77 FITS frames (51 MB), WASP-36150218062112.FITS → WASP-36150218100912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-150218.log (1004c317f5f5...), FinalLightCurve_WASP-36 b_2015-02-17.png (67fdf4da875a...), FinalLightCurve_WASP-36 b_2015-02-17.csv (c0034ceef172...)

Compute resources

Wall time 245.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 01:16Z.